6 Fig. 6.1 shows two bar magnets.



Fig. 6.1

(a) On Fig. 6.1, sketch the pattern and the direction of the magnetic field lines between the bar magnets. [2]

(b) Fig. 6.2 shows the same bar magnets with a coil of wire between them.

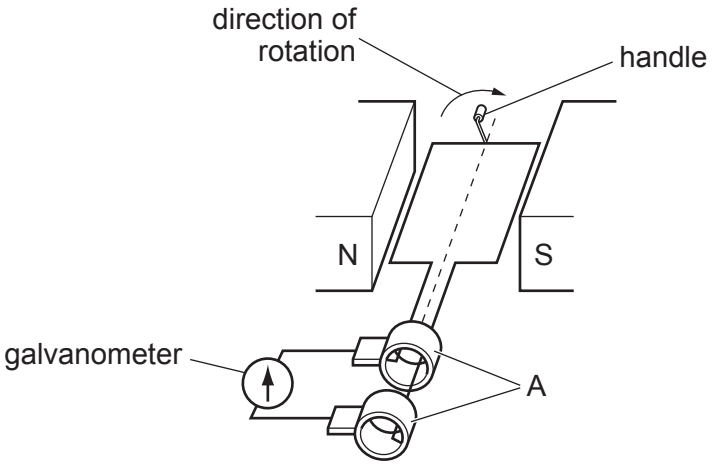


Fig. 6.2

(i) Name the parts labelled A in Fig. 6.2.

..... [1]

(ii) The coil of wire is rotated in the direction shown in Fig. 6.2.

On Fig. 6.2, draw an arrow to show the direction of the current in the coil. Explain your answer.

.....

..... [2]

(iii) Explain how rotating the coil in Fig. 6.2 continuously causes the galvanometer needle to show an alternating current.

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..... [4]

[Total: 9]

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